

Attention Is All You Need

Abstract

We propose the Transformer, a model based on attention. It uses softmax, matrix multiplication and gradient descent.

1 Introduction

Recurrent neural networks and backpropagation are standard for sequence modelling. We use attention and softmax instead.

2 Method

Scaled dot-product attention computes $\text{softmax}(QK^T / \sqrt{d}) V$ using matrix multiplication.

3 Experiments

We train with gradient descent on a translation task.